

> Retrieval Parameters

▼ ⓘ LLM Provider & Configuration

LLM Provider ⓘ

☐ Ollama (Local) ☒ AWS Bedrock

AWS Bedrock — Converse API

🔒 Auth via `aws configure` — no credentials needed here.

Bedrock Model ⓘ

Amazon Nova Lite (fast · cheap) ▼

ID `amazon.nova-lite-v1:0`

AWS Region ⓘ

us-east-1

🔧 Test Bedrock Connection

> 🧠 Embedding Configuration

🚀 Run Pipeline

🗑️ Clear Results

🧠 RAG Evaluation Dashboard

Compare retrieval-augmented generation against traditional summarization with comprehensive automated metrics

📄 Document Overview



2

PAGES



0

TABLES



0

IMAGES



800

WORDS



5

CHUNKS

▼ 📄 Extracted Text Preview

Showing 1,000 of 4,362 characters

1000

Many people want to have more money in the future, but they are not sure how to make that happen. The good news is that saving money and investing it wisely can help almost anyone build wealth over time. You do not need to be rich to start. You just need to understand a few basic ideas and be patient.

The first idea is saving. Saving means putting some of your money aside instead of spending all of it. If you earn money from a job, a good habit is to save a small part of it every month. Even saving ten or twenty dollars a week can add up to a large amount over many years. The key is to be consistent. Save regularly, no matter how small the amount.

The second idea is a bank account. When you put your savings in a bank, the bank pays you a small amount called interest. Interest is extra money the bank gives you for keeping your money there. For example, if you put one hundred dollars in a savings account and the bank gives you

No tables detected in the document.

Chunking Summary

Total Chunks

5

Text Chunks

5

Table Chunks

0

> [View Sample Chunks](#)



Retrieval Analysis

Total Chunks

5

Text Chunks

5

Table Chunks

0

Avg Chunk Length

994 c...

> 📄 All Chunks



Retrieved Contexts

Contexts Retrieved

3

Avg Distance

1.0000

Avg Similarity

1.2%

Retrieval Similarity Scores



Retrieved Context Details

▼ Context 1 · Similarity: 3.5% · Distance: 0.9652

of making mistakes. Everyone makes small financial mistakes when they are learning. The important thing is to learn from them and keep going. Read books, talk to trusted people, and take it one step at a time. Building wealth is not something that happens overnight. It is a slow and steady process that rewards those who are consistent, patient, and disciplined. To sum up, saving regularly, putting your money in a bank or the stock market, spreading it across many investments, and giving it time to grow are the four most important steps to building wealth. Anyone can do this. You do not need a lot of money, a special education, or a financial advisor. You just need to start.

> Context 2 · Similarity: 0.0% · Distance: 1.0062

> Context 3 · Similarity: 0.0% · Distance: 1.0284



Summary Comparison

SUMMARIZATION TASK

Summarize the document in exactly 3 lines.

Constraint: 3 lines

 RAG Latency

1.96s

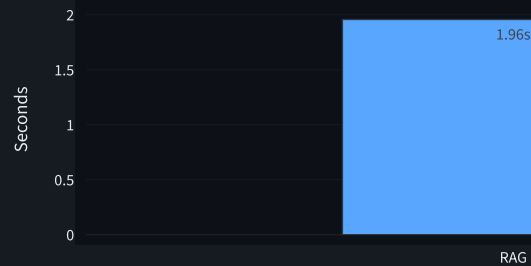
 Non-RAG Latency

1.99s

 Faster

RAG

Generation Latency



RAG Summary

⌚ 1.96s · 📄 46 words · 📄 1 lines · contexts: 3

Read Summary

Everyone can build wealth through consistent saving and wise investment. Saving a portion of your income regularly is key, even small amounts add up over time. Investing in a bank account or the stock market can further grow your savings, though the latter carries some risk.



Non-RAG Summary

⌚ 1.99s · 📄 33 words · 📄 3 lines · input words: 800

Read Summary

Saving regularly is the first step to building wealth. Putting money in a bank or the stock market helps it grow. Time and spreading investments across different companies are key to long-term growth.



Your Reference Summary (Ground Truth)

> Saving consistently and investing in the stock market are two practical ways for anyone to grow their money over time.

> Spreading investments across many companies — such as through an index fund — reduces the risk of losing money if one company performs poorly.

> Starting early is the most powerful advantage an investor can have, because time allows compound growth to multiply even small contributions into significant wealth.

```
</div>  
</div>
```

✅ RAG Advantages

- Grounded in retrieved chunks only
- Lower hallucination risk
- Traceable to source passages
- Stays within document scope

⚠️ Non-RAG Limitations

- Uses the entire document text
- May blend irrelevant sections
- Higher hallucination risk
- Harder to verify claims



Evaluation Metrics

📄 Lexical Metrics

ROUGE-1 (RAG) ⓘ

ROUGE-2 (RAG) ⓘ

ROUGE-L (RAG) ⓘ

BLEU (RAG) ⓘ

0.375

↓ -0.025 vs Non-RAG

0.091

↓ -0.011 vs Non-RAG

0.179

↓ -0.061 vs Non-RAG

0.000

↑ +0.000 vs Non-RAG

🧠 Semantic Metrics

BERTScore F1 (RAG) ?

0.881

↓ -0.009 vs Non-RAG

Cosine Similarity (RAG) ?

N/A

Faithfulness (RAG) ?

1.000

🔍 RAGAS Context Metrics (RAG-only)

Answer Relevancy ?

0.286

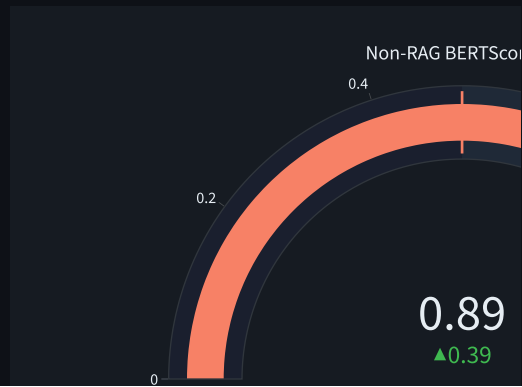
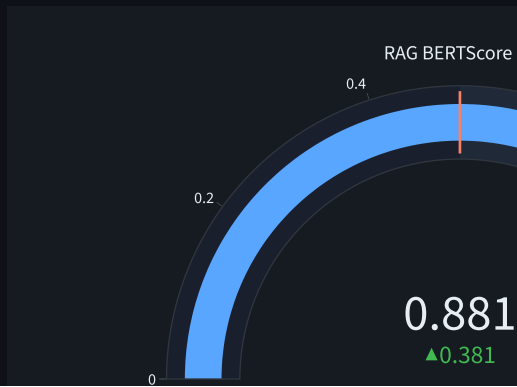
Context Precision ?

0.800

Context Recall ?

0.850

🧠 Semantic Similarity — BERTScore F1



RAGAS Metrics Radar (RAG-Specific)



Metric Analysis: Why Traditional Metrics Fall Short

Metric	What it Measures	Limitation
ROUGE-1/2/L	Word/n-gram overlap	High scores for domain vocab even if facts are wrong
BLEU	Precision of n-grams	Designed for translation; penalises paraphrasing
BERTScore	Semantic similarity	Similar meaning $\neq$ factual correctness
RAGAS Faithfulness	Context grounding	✅ Most reliable — checks claim support
RAGAS Answer Relevancy	Question relevance	Measures topical match, not accuracy
RAGAS Context Metrics	Retrieval quality	Evaluates retrieval, not generation

Research Question: "Can current automated metrics reliably distinguish between a 'correct' answer and a 'plausible but wrong' answer?"

Answer: NO — Traditional metrics (ROUGE, BLEU, BERTScore) measure surface-level similarity, not factual grounding. Only **RAGAS Faithfulness** directly checks whether the answer is supported by the retrieved context.



Visual Analytics

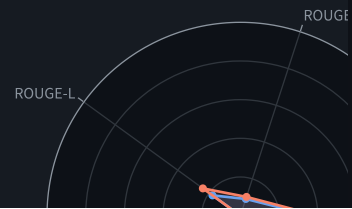


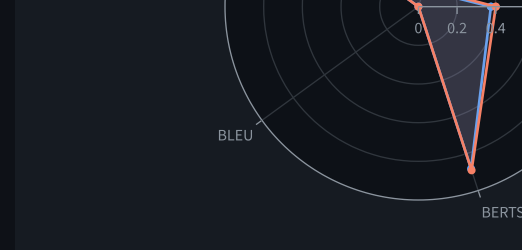
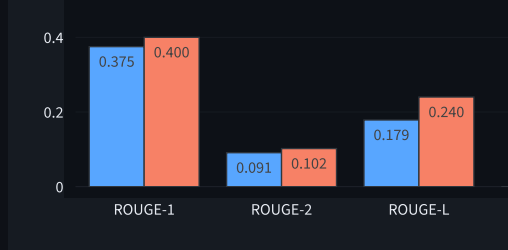
Metric Comparison — RAG vs Non-RAG

1
0.8
0.6

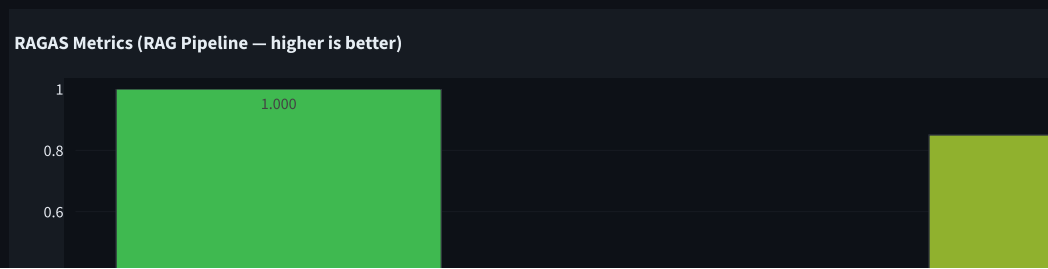


Performance Profile





RAGAS Metrics Detail





Results & Export



Final Evaluation Verdict

Task: Summarize the document in exactly 3 lines.

Constraint: 3 lines

Non-RAG outperforms on average lexical + semantic metrics by **0.021** points (RAG avg: 0.305 · Non-RAG avg: 0.326).

RAGAS Faithfulness is ****1.000****, indicating the RAG summary is well grounded in the retrieved context.



Document Stats

- Pages: 2
- Tables: 0
- Words: 800
- Chunks: 5



Performance

- RAG latency: 1.96s
- Non-RAG latency: 1.99s



Best Scores (RAG)

- ROUGE-1: 0.375
- BERTScore: 0.881
- Faithfulness: 1.000

Export Results

 [Download JSON](#)

 [Download CSV](#)

 [Download Report \(TXT\)](#)

>  [Full JSON Results Preview](#)